
Authority Is Not Recency: Validity-State Memory for Conversational Agents

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Abstract

1 Conversational memory systems decide what an agent believes about its user, and
2 most decide it with a proxy for authority: a recency boost, a date filter, or “take
3 the most recent value.” Recall Lab instead makes authority explicit — user claims
4 become traces carrying validity state (*active*, *superseded*), superseded traces are
5 rendered under *Past*, *no longer current* rather than deleted, and only user turns that
6 *set a value* may grant authority. Against seven baselines, including a full retrieval
7 stack, date-metadata filtering, a raw episodic read-time judge and a deterministic
8 max(timestamp) resolver, we report three findings. Where every correction is an
9 explicit value-set, validity state is *unnecessary*: keep-everything, latest-wins and
10 validity state all tie at 1.00, while every retrieval baseline fails the oldest super-
11 seded fact. What decides whether retrieval recovers such a chain is not retrieval
12 quality, recency weight, context size or visible timestamps, but *presentation or-*
13 *der*: re-sorting the same snippets chronologically moves first-fact accuracy from
14 0/5 to 4/5. And on an adversarial scenario containing a value-less cancellation and
15 a value-less re-mention, only validity state reaches 1.00, while the timestamp sort
16 falls to 0.40 and the strongest retrieval stack to 0.72. The principle: a statement
17 should change memory only if it sets a value. We also report a failure our own
18 consolidation introduced, and how tracing located it.

19 1 Introduction

20 An assistant that remembers its user must answer a question retrieval does not pose: *of the things*
21 *this user has told me, which are still true?*

22 The field largely answers with recency. Vector stores add recency boosts; memory layers filter
23 by date added; agent frameworks resolve conflicts by taking the latest value. These are proxies
24 for **authority** — being the current, user-sanctioned truth — and they work while every correction
25 explicitly restates a value. They break in two places, both common in real conversation:

- 26 • **Value-less cancellation.** “*Cancel that change, leave it as it was.*” The most recent state-
27 ment about the attribute is the one being cancelled; a timestamp sort returns the cancelled
28 value.
- 29 • **Value-less re-mention.** “*Green is still such a beautiful color, I always come back to it.*”
30 The user is reminiscing, not setting a preference; a timestamp sort lifts a superseded value
31 back to current.

32 Both share a structure: **the most recent mention of an attribute is not an assertion of its current**
33 **value.** Recency reads *when* something was said, not *whether the user was setting it*.

Contributions: (i) a memory architecture making authority explicit, where forgetting removes authority rather than history; (ii) a controlled comparison against seven baselines, including the two strongest non-retrieval alternatives usually omitted — keep-everything and deterministic latest-value; (iii) **a pre-registered negative result we report rather than bury**, naming the scenario property under which validity state is unnecessary; (iv) isolation of presentation order as the decisive variable for retrieval-based memory; (v) an adversarial scenario on which only validity state holds.

2 Related work

Consolidation can degrade memory. Zhang et al. [2026] show that LLM-driven consolidation degrades over time — utility rises, then falls below a no-memory baseline — and that an episodic-only control retaining raw trajectories stays competitive with every consolidator they test. We therefore make `episodic_judge` a first-class baseline, and Section 5 reports that our own consolidator reproduced their finding before we fixed it: pre-guard, the brief scored 0.64 against the episodic control’s 0.92. Our claim is not that consolidation is safe, but that a nameable rule about *what may be consolidated* separates the two regimes.

Memory systems and benchmarks. Production memory layers extract and consolidate salient facts into a persistent store [Chhikara et al., 2025]; Du [2026] surveys the mechanism families, naming continual consolidation and learned forgetting as open problems. Superseded facts are already a recognised failure axis: LongMemEval [Wu et al., 2024] carries a *knowledge updates* task category, and LoCoMo [Maharana et al., 2024] evaluates very long-term multi-session dialogue. Those benchmarks measure how often systems fail; we hold retrieval fixed and vary only how authority is decided, to isolate *which* mechanism is responsible.

Position and order effects. Liu et al. [2024] show that models use information mid-context less reliably than at the edges. Our Section 4.2 result is adjacent but distinct: neither a larger context nor visible timestamps recovered the missed fact — only re-sorting the *same* snippets chronologically did, an ordering effect over and above positional degradation. Finally, Recall Lab adapts truth maintenance [Doyle, 1979] and belief revision [Alchourrón et al., 1985] to the LLM agent memory setting rather than claiming to invent validity tracking.

3 Method

Recall Lab maintains a raw **episodic log** (evidence), a **trace store** (what consolidation reasons over) and a **consolidated brief** (what the agent reads). A **sleep job** runs after each simulated day: score the *user turn only* for salience, so the system cannot grant authority to its own output; compress high-salience claims into traces; check each against active traces and confirm, correct or add; mark corrected traces *superseded* rather than deleting them; render active traces as current memory and superseded ones under *Past, no longer current* with lineage labels. Separating those sections lets the agent answer both “where do you ship now?” and “where did you ship before?”

The value-setting guard. Following the failure in Section 5, salience gates on whether the user turn *assigns the attribute as a current value*; a sentiment or reminiscence mention is UNRELATED, not CORRECTION, with a secondary guard at the contradiction classifier. This is a general rule about intent, not a scenario-specific patch. A brief-invariant check runs after every pass, reporting violations into the pass summary.

Baselines. `sliding_window_2` (last N turns); `budgeted_window` (window sized to Recall Lab’s token budget, so shorter prompts cannot win on length alone); `vector_topk_5` (flat top- k similarity); `strong_rag` (query rewriting $\rightarrow$ hybrid dense + BM25 $\rightarrow$ reciprocal rank fusion [Cormack et al., 2009], $k=60 \rightarrow$ recency boost $\rightarrow$ LLM reranking; no validity state by design); `strong_rag_dated` (adds `added_at` timestamps, timestamp recency, and a metadata filter, `recency_window_days=1.5`); `episodic_judge` (inject the whole log every turn, decide currency at read time); `deterministic` (an LLM extracts value-setting (attribute, value) pairs; the current value is $\max(\text{added_at}, \text{turn})$ in pure Python, no model judgement of currency). The last makes the sharpest objection to our thesis concrete: *if the current answer is always the most recently stated value, why classify contradictions at all?*

Table 1: Recall accuracy, 5 seeds each, pinned provider. Read the deterministic row against the last: a timestamp sort *ties* the validity brief where corrections are explicit value-sets, and collapses where they set no value. *Coverage artifact: a two-turn window spans the relevant turn for that one question.

Condition	relocation_chain	correction_intent		
	Acc.	Acc.	Revert	Re-assertion
sliding_window_2	0.00	0.20	0/5	5/5*
budgeted_window	0.08	—	—	—
strong_rag_dated	0.40	0.36	0/5	3/5
vector_topk_5	0.48	0.68	0/5	3/5
strong_rag	0.76	0.72	0/5	4/5
episodic_judge	1.00	0.92	3/5	5/5
deterministic	1.00	0.40	0/5	0/5
recall_lab_brief	1.00	1.00	5/5	5/5

Scenarios. Both are 4-day customer-assistant conversations, one persona, ending in a 5-question recall evaluation. `relocation_chain`: the shipping city changes across three cities and two facts never change; every change is an explicit value-set. `correction_intent`: adversarial, built after the Section 4.1 result, adding a *stale re-assertion* (colour green→blue, then green fondly re-mentioned without changing the preference; expect blue), a *revert* (Berlin→Munich, then the change cancelled without restating Berlin; expect Berlin), and an *implicit correction by negation*, plus a confirmation control and a history question.

Measurement. Recall accuracy graded by an LLM judge with a 3-call audit; 0 of 175 graded answers split in the Section 4.3 campaign. Model calls are provider-pinned with fallbacks off, because non-deterministic provider routing adds variance unrelated to the memory architecture. 5 seeds per condition per scenario.

4 Results

4.1 Where corrections are explicit, validity state is unnecessary

`strong_rag` per-question: current city 4/5, previous 5/5, **first city 0/5**, both stable facts 5/5. A full industry stack recovers the *recent* end of a correction chain and collapses on the *oldest* superseded fact: recency approximates authority near the present and decays into the past.

But **three very different strategies tie at the ceiling**. Validity state is not justified here: every change is an explicit user-stated value update, exactly where “take the latest” is correct by construction. We pre-registered this as a falsification target and report it. Only cost separates them: 438 mean input tokens/turn against `episodic_judge`’s 974, a gap that widens with conversation length.

Date filtering is actively harmful. `strong_rag_dated` (0.40) buys a perfect current-city answer by deleting every fact older than the window, including a stable restriction stated on day 1 that never changed (0/4): a hard filter cannot tell “old but still true” from “old and superseded.”

4.2 The decisive variable is presentation order

To remove the objection that the first-city failure is an artifact of two hyperparameters, we ablate recency weight and top-*k*, then give the model the entire conversation with timestamps visible, in relevance order and in chronological order. Real BM25 hybrid retrieval throughout.

Two pre-registered predictions failed. First-city accuracy did *not* rise with top-*k*: `kFULL` stayed 0/5, so the fact was in context and still missed — never a coverage problem. Nor did full context with visible timestamps approach 1.00 in relevance order (0.80, 0/5).

Same retrieval, same timestamps, same context size: re-sorting oldest-first took first-city from 0/5 to 4/5. Relevance ranking scrambles chronology and the model does not reliably re-sort by date on its own. Chronological presentation fixes this at keep-everything token cost; the validity brief reaches the same accuracy at 45% of the tokens, because it stores ordered lineage explicitly rather than reconstructing it per query.

Table 2: Fair RAG sweep, `relocation_chain`, 5 seeds. Same retrieval and timestamps throughout; only presentation order differs between the last two rows.

Config	Mean acc.	First-city	Tokens
<code>strong_rw0.3_k5</code>	0.68	0/5	429
<code>strong_rw0.3_kFULL</code>	0.76	0/5	902
<code>fairshot</code> (relevance order)	0.80	0/5	969
<code>fairshot_chrono</code> (time order)	0.96	4/5	979
<i>ref: recall_lab_brief</i>	1.00	5/5	438

120 4.3 Where corrections set no value, validity state is necessary and sufficient

121 The timestamp sort that tied at 1.00 collapses to 0.40; only the validity brief holds. On the re-
 122 vert every retrieval baseline scores 0/5 — retrieval cannot represent a cancellation at all — and
 123 deterministic returns Munich exactly as predicted by construction. Reading the whole raw log
 124 does not force a single authority decision either (`episodic_judge` 3/5). Both discriminators share
 125 one property: **the most recent mention sets no value.**

126 5 Consolidation can promote the wrong thing

127 Before the guard, `recall_lab_brief` scored **0.64** on `correction_intent`, *below* `strong_rag`
 128 (0.80) and `episodic_judge` (0.92), and 0/5 on the stale re-assertion it was designed to catch.

129 Trace analysis over the full export (5,797 model calls) corrected our initial diagnosis. We first logged
 130 it as a contradiction-classifier bug; the traces showed it begins one layer earlier. The **salience judge**
 131 scored the reminiscence line at 0.55, reasoning “*User expresses a stable aesthetic preference for the*
 132 *color green,*” and promoted it as a preference. The classifier then received “favorite colour is green”
 133 against active “blue” and returned CORRECTION — locally correct, given a wrong input. Root cause:
 134 salience promoting sentiment as a value-set.

135 The fix moved upstream. Post-fix: blue 0/5→5/5; Berlin 5/5→5/5, so the guard did not over-
 136 suppress real changes; the history question 1/5→5/5 once the lineage stopped inverting. Baselines
 137 were unchanged, confirming the scenario did not get easier — Recall Lab moved.

138 This is the real cost of consolidation that Zhang et al. [2026] warn of: a system that decides what
 139 to remember can decide wrongly in a way keep-everything cannot. The mitigation is a general rule
 140 about value-setting, and the failure was findable only because every model call was traced.

141 6 Limitations

142 Two synthetic scenarios, one persona, 5 seeds per condition; the pre-registered 30-conversation cam-
 143 paign has not been run. One agent model, so we cannot separate model-specific effects, particularly
 144 the Section 4.2 finding that the model does not re-sort scrambled chronology — a stronger reasoner
 145 may. LLM-as-judge grading, audited with 3 calls but not validated against human annotation. No
 146 statistical testing; spreads are reported instead of the planned Wilcoxon and bootstrap CIs. Criti-
 147 cally, `correction_intent` was authored by us after `relocation_chain` failed to discriminate;
 148 independently authored adversarial scenarios are needed.

149 7 Conclusion

150 Recency is a proxy for authority that holds while corrections are explicit value-sets and breaks where
 151 the most recent mention of an attribute assigns no value to it: cancellations and reminiscences.
 152 Making validity state explicit resolves both at bounded token cost, where a retrieval stack, a date
 153 filter, keep-everything and a latest-value resolver each fail at least one. Forgetting should remove
 154 authority, not erase history; authority should be granted by intent, not timestamp. Code, scenarios,
 155 per-run reports and a dated log of every pre-registration and its outcome — including the predictions
 156 we got wrong — are available anonymised on request.

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